Kushal Sharma

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SUMMARY

Senior Software Engineer with 7+ years of experience architecting high-throughput distributed backend systems, enterprise data architectures, and microservices using Java, Python, Spring Framework, and Kafka, with specialized expertise in embedding production GenAI capabilities, RAG pipelines, AI features into applications and resolving complex cloud system issues. Proven track record in enterprise integrations, bridging traditional enterprise infrastructure with modern AI integration patterns including AI-assisted workflows and tooling automation (Spring AI, PyTorch, Vector Search, MCP). Dedicated to delivering scalable, secure, and resilient software across finance and enterprise software applications.

EDUCATION

University of Florida - Master of Science, **Computer Science** **May 2023**
University of Mumbai - Bachelor of Engineering, **Computer Engineering** **May 2017**

WORK EXPERIENCE

Wissen Technology | *Senior Software Developer* | *India* **May 2026 - October 2026**

- Implemented **Python**-based data reconciliation pipeline to pull and compare relational data across two different **database sources**, checking data accuracy for downstream enterprise reporting.
- Implemented **REST API** for bulk data imports that dynamically generates Excel templates, helping speed up the data ingestion process.
- Strengthened **system security** and **operational compliance** by actively auditing the codebase and deploying targeted patches to resolve **critical vulnerabilities** and **technical hygiene** items.

Remora | *Back End Developer* | *United States* **July 2023 - December 2025**

- Played a key role in launching the platform's next **major version**, delivering continuous, incremental feature updates helping grow the **enterprise client base** by **50%** (from 80 to 120).
- Added **GenAI** features and a **RAG chatbot** into the **Java backend** using **Spring AI** and **vector search**, which improved query relevance and user engagement.
- Integrated **OAuth2** authentication (Google and Apple) into a server-side rendered application to securely manage user sign-ins and store identity data in the database.
- Built an audit system for the main portal using **Spring Security** and **MongoDB** to track user actions, and paired it with an admin dashboard featuring a **JSON** diffing tool so admin could easily filter and compare what entity modifications were made.
- Developed **REST APIs** and **microservices** using **Java**, **Spring Boot**, and **Kafka**, managing rapid application expansion by containerizing both the core backend and AI-driven workflows on Kubernetes, improving overall system performance by **30%**.
- Collaborated cross-functionally with front-end, data science, and product teams to integrate **ML-powered** features and **AI-driven** workflows into production systems using **Spring AI** and **micro service** orchestration using **Kubernetes (K8s)**.
- Mentored junior developers on best practices focusing on **Spring Security, Test-Driven Development (JUnit/Mockito)** for code quality.

Itiviti | *Full Stack Engineer* | India **October 2020 - July 2021**

- Built an automated daily **log parser** to extract specific **financial metrics** into the database, and created a corresponding UI to give users easy, filtered access to the extracted data.
- **Optimized persistence layers** by designing complex **MongoDB** aggregation pipelines and implementing **LRU caching in Java**, reducing log-scheduler query latency by 7%.
- **Maintained 70%+ unit test coverage** using **JUnit, Mockito**, and **Jacoco** ensuring the stability of critical financial transaction modules.

QualityKiosk Technologies Private Limited | *Software Developer* | India **July 2019 – February 2020**

- Implemented a scalable **Spring Boot** backend integrated with **Elasticsearch (ELK Stack)** to aggregate **logs and metrics**, inferring analytical data for an **AngularJS** dashboard.
- Implemented robust **security protocols** using **Spring Security (JWT)** to protect application from unauthorized users.
- Audited internal software with **CAST** Software to pinpoint **security** gaps, efficiency issues, and **technical debt** across the codebase.
- Designed and implemented full-stack web application features using **Java**, **Spring Framework**, **Hibernate**, and **AngularJS**, while utilizing **JHipster** to scaffold utility services and **Liquibase** for database change management.

Verdantis Tech Private Limited | *Software Developer* | India **July 2017 - June 2019**

- Architected a secure data integration layer using **Java, JSch**, and **SFTP/Web Services** to process **ZIP/XML** payloads, enabling seamless data synchronization increasing efficiency by **300%.**
- Developed a custom **Java** and **MySQL** backend to automate L1 duplicate material detection and error reporting, eliminating manual efforts.
- Configured **UTF-8** encoding across DB tables and application endpoints to fix broken text formatting and support non-English languages.
- Streamlined the release lifecycle by building automated **Java** utilities for **XML** packet generation, reducing deployment friction between development and testing teams.

INTERNSHIP

University of Florida | *Teaching Assistant* | *United States* **January 2023 - May 2023**

- Mentored graduate students in the implementation of core machine learning algorithms and guided technical data mining projects under Professor Ruogu Fang.

Sonos | *SDE Intern* | *United States* **June 2022 - August 2022**

- Designed and implemented real-time distributed tracking systems using **Spring Boot**, **Spring JPA**, and **ReactJS**, improving event visibility by **90%**.
- Leveraged **asynchronous programming and schedulers** to optimize cross-device synchronization.
- Achieved **90% code coverage** using **TDD (JUnit, Mockito, Jacoco)**, strengthening system reliability.
- Deployed a multithreaded peer-to-peer system using RMI and AWS EC2 for high concurrency and reliable distributed processing.

A3Pixels | *Web Developer Intern* | *India* **December 2015 – January 2016**

- Engineered foundational full-stack **web components** and developed interactive e-learning modules utilizing **Adobe Captivate.**

PROJECTS

Enterprise RAG & LLM Fine-Tuning Engine | Python, PyTorch, Java, Spring Boot

- Architected a scalable enterprise AI platform combining custom PyTorch LLM fine-tuning and RAG pipelines deployed via high-throughput Spring Boot microservices.

Twitter Clone | F#, REST APIs

- Built a distributed social system using the Actor Model and REST APIs; verified fault tolerance through integration tests.

Compiler | Java, JUnit

- Developed a lexer, compiler, Abstract Syntax Tree, Type Check Visitor, and Bytecode Generator using Java and test cases using JUnit .

P2P Distributed Architecture | F#

- Developed a multithreaded peer-to-peer system to find the next patterned number.

Inventory Management | Java, Microservices, REST, LRU Caching

- Developed a fault-tolerant inventory system using microservices and node replication to improve latency and handle node failures.

SKILLS

Languages: Java, Python, PHP, JavaScript, TypeScript, React, Next.js, HTML5/CSS3, SQL.

AI & Agentic Systems: MCP (Model Context Protocol), RAG (Retrieval-Augmented Generation), Vector Search, AI-Enabled SDLC, LLM Fine-Tuning, Parallel Training Pipelines, PyTorch, CUDA.

Frameworks: Spring Boot, Spring JPA, Spring Security, Fastify, Hibernate, Laravel, React, Node.js, Angular, Kafka

Databases: MySQL, PostgreSQL, MongoDB, Cassandra

Cloud & DevOps: AWS (S3, SQS, Lambda), Docker, Kubernetes, Jenkins, Git, Nginx, GitHub Actions.

Tools & Testing: JUnit, Mockito, Maven, Gradle, Redis, ELK Stack (Elasticsearch), Swagger

Concepts: Distributed Systems, Microservices, RESTful API Design, System Design, CI/CD, Agile, Root Cause Analysis (RCA), Data Pipelines, Cloud-Native Architecture, Event-Driven Architecture (Kafka, RabbitMQ), OAuth, SSO, JWT, Role-Based Access Control (RBAC).